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INTERPRETABLE MUSIC HARMONIC ANALYSIS THROUGH MULTILINEAR MIXTURE OF EXPERTS

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ABSTRACT

Transformer-based models achieve state-of-the-art performance in music harmonic analysis, and more specifically in Roman Numeral Analysis (RNA), a fundamental task in music information research that assigns harmonic function labels to musical scores. However, current models lack interpretability, limiting their utility for musicological research where understanding the analytical results is crucial. In this paper, we introduce μMoE-RNBERT, the first interpretable deep learning system for RNA, which integrates multilinear mixture of experts (μMoE) into the RNBERT architecture. Our approach maintains near-identical parameter count and comparable performance while enabling expert specialization in musicologically meaningful patterns. Through quantitative analysis and visualization of expert activations, we demonstrate that individual experts learn to specialize in harmonic contexts following music-theoretical principles, including dominant-to-tonic relationships and circle-of-fifths patterns across key changes.

***Index Terms*—** Roman numeral analysis, mixture of experts, music information retrieval, music harmonic analysis

1. INTRODUCTION

Roman numeral analysis (RNA) [1],[2] is one of the main tasks of Western music harmonic analysis, assigning chord labels (e.g., ii-V-I) to each note segment of a score based on the current key and indicating their role. Deep learning approaches have emerged for automatic roman numeral analysis of symbolic music, either fine-tuning pre-trained models [3] or training models on labeled data from scratch [4],[5],[6],[7]. Among these approaches, models such as recurrent networks [5],[8], [9], transformers [6], and graph neural networks [4],[7] have achieved increasingly strong performance.

RNBERT currently achieves state-of-the-art performance [3] in the task. This model fine-tunes MusicBERT [10], a BERT-like encoder-only transformer pre-trained on symbolic music data. However, existing architectures lack interpretability [11], [12], providing predictions without any explanation of their underlying reasoning process. This lack of interpretability poses significant challenges for AI-driven musicological research. While these models may correctly identify fundamental harmonic progressions like dominant-to-tonic (V-I) relationships, it is unclear whether the model has learned genuine music-theoretical principles, such as voice leading patterns or functional harmony relationships, or simply memorized statistical correlations in the training data. Without this understanding, musicologists cannot validate the model's theoretical soundness, trust its analyses for scholarly work, or gain new insights into harmonic structure that could advance music theory itself.

To address this limitation, we introduce μMoE-RNBERT[1], an interpretable variant that maintains high performance while providing insights into its harmonic reasoning. Our approach integrates multilinear mixture of experts (μMoE) [13] by replacing the MLP layers within RNBERT's feed-forward blocks with μMoE layers. The μMoE blocks offer advantages over both standard dense layers and traditional sparse MoE architectures [14], allowing different experts to specialize in distinct harmonic patterns and musical contexts while maintaining computational efficiency. The contributions of this paper are as follows:

- We propose, to the best of our knowledge, the first intrinsic interpretable deep learning system for roman numeral analysis. Unlike post-hoc explanations, our model offers interpretability by design, enabling musicologists to understand and validate AI-driven harmonic analyses.
- We introduce μMoE-RNBERT, which achieves comparable performance to the original RNBERT using the same number of parameters while providing interpretable insights into its harmonic analysis decisions.
- We demonstrate that individual experts learn musically meaningful specializations, including key-specific patterns following the circle of fifths and dominant-to-tonic relationships, validated through both quantitative analysis and musicological evaluation on the same dataset as [3]. This large corpus consists of different works, including different corpora from Digital and Cognitive Musicology Lab like [15], [16], [17], the TAVERN set with works from Beethoven and Mozart [18], the set of Beethoven Piano Sonatas first movements from [8], and lastly the various scores from When in Rome meta-corpus [19].

2. RELATED WORK

2.1. Automated Music Harmonic Analysis

Automated music harmonic analysis, particularly Roman numeral analysis, represents a growing area of research driven by advances in deep learning. Chen et al. [8] studied RNA under a multi-task learning (MTL) framework [20], outperforming models trained for individual tasks separately. Transformer-based approaches have also shown promise, with the Harmony Transformer (HT) [6] demonstrating strong performance in chord boundary identification. AugmentedNet [5], [9] further improved RNA accuracy by incorporating additional tonal tasks within an MTL framework. Recent work

[1]Code available at: https://github.com/TomusD/muMoE-RNBERT

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has explored novel data representations, including ChordGNN [4], which represents musical scores as graphs where notes are nodes and their relationships form edges. This approach was extended with the MusGConv block [7], designed to capture human music perception through relative pitch and rhythmic relationships between notes. Despite achieving high accuracy, these models operate without interpretability of their results.

The current state-of-the-art approach, RNBERT [3], fine-tunes MusicBERT [10], a large-scale transformer pre-trained on over one million MIDI files through masked language modeling. RNBERT frames RNA as a multi-task token classification problem, predicting key, quality, inversion, and degree for each note token. While this approach achieves superior performance compared to previous methods, the interpretability limitation persists, leaving the model's harmonic reasoning processes opaque to musicologists.

2.2. Mixture of Experts and Interpretability

Mixture of Experts (MoE) [14] architectures use conditional computation to activate different model components based on input characteristics, enabling expert specialization in specific subtasks. This specialization can enhance model interpretability by creating more modular internal representations. MoE has enabled scaling to trillions of parameters [21], [22] while maintaining computational efficiency. Predominant sparse MoE approaches [23] can suffer from training instability and expert underutilization. Recent work on multilinear MoE [13] addresses these limitations through factorized weight representations, enabling fine-grained expert specialization and efficient scaling, making it our chosen method. µMoE yielded experts who learned meaningful patterns such as object-specific features in computer vision and syntactic/semantic concepts in natural language processing.

Recent transparent approaches in MIR include explainable descriptors to estimate music difficulty [24], and model-based deep learning, which integrates domain knowledge such as physical or musicological models, directly into network architectures to improve interpretability and controllability [25]. Furthermore, post-hoc methods have been modified for the audio domain, for example, audioLIME generates listenable explanations by switching on and off source-separated components [26], while MusicLIME extends this to multimodal settings to interpret interactions between audio and lyrical features [27]. We chose an intrinsic interpretability method like µMoE over post-hoc alternatives for several reasons. Post-hoc methods may not recover the original model's computations fully, and they introduce additional runtime costs. More importantly, they typically identify features at the neuron level. In contrast, the conditional computation in µMoE allows us to discover entire subnetworks that correspond to specific musical tasks and patterns.

Despite these advances, mixture of experts architectures have not been applied to music analysis, where interpretability is particularly crucial for musicological validation. Current harmonic analysis systems operate without explaining their reasoning, limiting their scholarly utility. We address this dual gap by introducing the first interpretable harmonic analysis system through multilinear mixture of experts, combining the scalability of MoE with the theoretical needs of music information research.

3. PROPOSED METHOD

In this section, we develop an interpretable Roman numeral analysis system through multilinear mixture of experts. We first describe the baseline RNBERT architecture and its training methodology, then introduce the multilinear mixture of experts model, and finally detail our integration of µMoE layers into RNBERT to develop the interpretable µMoE-RNBERT model.

3.1. Roman Numeral Analysis with RNBERT

Roman numeral analysis involves predicting four interconnected harmonic attributes for each note in a musical score: key (the tonal center), degree (scale position of the chord root), quality (major, minor, diminished, etc.), and inversion (bass note position).

RNBERT address this task as a multi-task token classification problem using the pre-trained MusicBERT encoder. The model employs MusicBERT's base architecture with 12 transformer layers, each containing multi-head attention followed by feed-forward MLP blocks, with a hidden dimension of 768. Musical scores are encoded using OctupleMIDI representation, where each note is represented as an octuple of features including pitch, duration, velocity, and metric position [10]. To ensure homophonic texture analysis, salami-slicing [28] creates discrete segments where each can be labeled with a single roman numeral. Scores are segmented into 1000-token sequences with 250-token overlap, quantized at the 64th note level, with duplicate notes sharing identical pitch and timing removed to reduce sequence length.

The training procedure freezes the initial 9 layers to prevent overfitting, fine-tuning only the top 3 layers. Data augmentation of the training set includes transposition to all chromatic keys and temporal scaling based on score duration relative to the training set mean. Classification heads consisting of two-layer MLPs are added for each task, with the overall loss combining cross-entropy losses across all prediction tasks. During inference, predictions from overlapping segments are merged through cross-fading of output logits.

3.2. Multilinear Mixture of Experts

Multilinear Mixture of Experts addresses limitations of traditional sparse MoE through factorized weight representations. Instead of discrete expert routing, µMoE represents weight tensors using low-rank decompositions such as Canonical Polyadic (CP) [29] or Tensor Ring (TR) [30], enabling differentiable expert combination.

For a two-layer MLP with input dimension I, hidden dimension H, and output dimension O, the µMoE transformation with N experts is:

$$y_t = \sum_{n_2=1}^{N} \sum_{h=1}^{H} w^{(2)}_{n_2 h} \cdot \text{GELU}\left(\sum_{n_1=1}^{N} \sum_{i=1}^{I} w^{(1)}_{n_1 i} \cdot z_{ti} \cdot a_{tn_1} \right)_h \cdot a_{tn_2}$$

where $z_t \in \mathbb{R}^I$ is the input vector for token t, $a_t \in \mathbb{R}^N$ are the expert coefficients computed by the gating network, and $W^{(1)} \in \mathbb{R}^{N \times I \times H}$, $W^{(2)} \in \mathbb{R}^{N \times H \times O}$ are the factorized weight tensors.

The gating network employs entmax [31] activation to compute expert coefficients, allowing sparse yet differentiable expert selection. The factorized representation enables scaling to thousands of experts while maintaining the same parameter count and computational complexity as a single dense MLP.

3.3. µMoE-RNBERT Architecture

We integrate µMoE into RNBERT by strategically replacing the MLP blocks within each transformer's feed-forward component, targeting the post-attention processing stage where harmonic pattern recognition likely occurs. Specifically, both linear layers of

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each feed-forward MLP are replaced with corresponding μMoE layers, allowing the model to develop specialized pathways for different harmonic contexts. Our implementation employs different Number(N) of experts with both factorization methods, while trying to maintain the RNBERT parameter count of 26.7M, with expert coefficients computed independently for each note token to enable context-dependent activation.

The choice to modify feed-forward layers rather than other components is motivated by several architectural considerations. The multi-task learning framework benefits from shared expert specialization occurring before task-specific processing, while the classification heads themselves have varying output dimensions across tasks (e.g., degree prediction requires 189 classes while inversion uses only 4), which would complicate unified expert interpretation. Additionally, roman numeral analysis involves complex post-processing steps that could obscure the direct relationship between expert activations and musical patterns.

This design preserves RNBERT's established training protocol and inference procedure while introducing interpretable expert specialization. The resulting architecture enables experts to develop distinct competencies in harmonic functions, key relationships, and musical patterns, providing unprecedented insight into the model's analytical reasoning process.

4. EXPERIMENTS

4.1. Experiment Setup and Implementation

To ensure our results are comparable and evaluate the μMoE-RNBERT architecture, we use the same dataset, train/validation/test split, the same fine-tuning protocol, and the same pre/post-processing from the original RNBERT [3], and as an evaluation metric, we use accuracy. For our quantitative evaluation, we use the occurrences of predicted labels above a threshold to count where and how much the experts activate the most, and for our qualitative evaluation, we use piano roll plots. To better interpret the musical patterns we also use the circle of fifths, which is a method of arranging the twelve pitch classes that can be described mathematically as elements of the set of integers modulo $12, \mathbb{Z}_{12} = \{\bar{0}, \bar{1}, \ldots, \overline{11}\}$ [32]. Our setup consists of a single NVIDIA RTX 3080 GPU with 10GB of VRAM, 32GB of DDR5 RAM, and a Ryzen 7 7700X 8-core CPU.

We use PyTorch for our implementations, and we fine-tune the top 3 layers of MusicBERT for 50,000 steps using a batch size of 4 and the Adam optimizer, with a learning rate of 2.5×10^{-4} and a linear warmup of 2,500 steps, followed by a linear decay to zero.

We use the implementation method of large language models for the μMoE layers, so we set the ranks to match the parameter count of the base RNBERT model, and we tune the number of experts, N, for both CP and TR methods. For the TR method, we also fix $R1 = R2 = 4$. With this parameterization, we manage to have similar parameter counts and FLOPs as a single, dense MLP block for N experts in each μMoE layer block [13].

4.2. Results

Table 1 shows the evaluation results on the test set for the base RNBERT model from [3] and our μMoE-RNBERT with 2 factorization methods from [13]. In μMoE-RNBERT trials, we show two variants of $N = 48$ and $N = 256$ experts. As we hypothesized, we have minimal diminishing returns with the addition of the explanatory nature we added to the model, which we will discuss in the next subsections. A two-proportion Z-test confirmed no statistically significant difference between our method and the baseline ($p = 0.80$), with a maximum Cohen's h of 0.033 indicating negligible effect size. Furthermore, the number of parameters is nearly identical to the original RNBERT, introducing no extra computational cost. With the trained model, we collect the coefficients at inference time from the test dataset, which consists of 146 scores.

Table 1. The accuracy results comparison of RNBERT with our model on the test dataset. Results are shown for different numbers of experts (N) and two factorization methods.

Models	Accuracy					
	Degree	***Quality***	***Inversion***	***Key***	***RN***	***Params***
RNBERT	**76.2**	**86.7**	**87.2**	82.2	**62.0**	26.7M
CPμMoE-RNBERT N=48	74.7	86.5	86.9	81.4	60.9	26.7M
TRμMoE-RNBERT N=48	74.6	86.3	86.8	81.6	60.4	26.7M
CPμMoE-RNBERT N=256	74.9	86.4	**87.2**	**82.3**	61.4	26.8M
TRμMoE-RNBERT N=256	74.7	86.2	87.1	82.1	61.0	26.2M

In the following analyses, we want to investigate and answer the question: "What musical pattern does each expert specialize in?" We begin with a Quantitative Analysis of the expert activation coefficients in the test set. After detailing our methodology for handling data from overlapping input segments using cosine similarity, we will statistically show that the experts learn to focus on the dominant-to-tonic (V-I) harmonic relationship on key-specific activations that follow the circle of fifths. Recognizing the data-related limitations of this statistical analysis, we then continue with the Qualitative Analysis to provide a deeper, musicological context. Using piano roll visualizations with activation heatmaps for different scores, we will showcase our findings, such as how experts respond to key modulations, the clustering of note activations based on the scale of the key, the differentiation of a key's harmonic function, and lastly state some findings of granular specializations in the model's deeper layers. These analyses provide an understanding of how our model develops an interpretable and musically consistent understanding of harmonic analysis.

4.3. Quantitative Analysis

Firstly, since 9 out of 12 layers are frozen, we have the option to examine the coefficients of each expert from the last three layers. Due to the segmentation within the octupleMIDI encoder, we have overlapping coefficients, which we investigate for redundant information. So, we compute cosine similarity for all the overlapping notes per score segment for N experts. The cosine similarity analysis confirms high redundancy between overlapping notes, with a mean similarity $\mu = 0.99$. This high similarity is consistent across all experts and segments, with a low standard deviation $\sigma = 0.013$. However, some overlaps show that in some cases the segments share only a minimum similarity of 0.689, suggesting instances with dissimilar information, albeit small in number. Based on those findings, we address the overlapping coefficients with two methods: i) drop all instances of the overlapping segments except the first, and ii) calculate the mean of the overlapping coefficients. Both methods showed similar results, although the simpler dropping method is computationally more efficient. For the rest of the analyses, we will use the mean method.

Fig. 1 displays the most active experts above the threshold ≥ 0.4, and for which roman numeral labels (a combination of key, degree, inversion, and quality) each expert activates the most. The roman nu-

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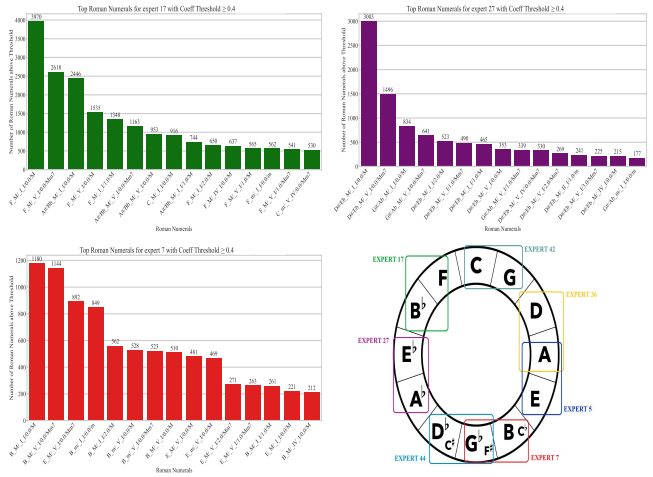


Fig. 1. Top-most activated roman numerals for the most active experts with a threshold ≥ 0.4 from layer 12. Roman numerals in the figure are labeled from the four harmonic attributes: key, degree, inversion, and quality in this order. In the bottom right corner, a circle of fifths with expert key specialization. Each box corresponds to the dominant-to-tonic relation that each expert is most active on, clockwise. **Zoom for clarity.**

merals with the higher activation number in each plot mostly create a dominant-to-tonic relation (V-I), with higher occurrences of I and V. We also observe that we have higher activations on major modes than minor ones. In the bottom right corner of Fig. 1, we see the circle of fifths keys pattern that each expert specializes in. The activation experts activate clockwise, for example, from A-flat major to E-flat major, and in most cases, we observe higher activation values in E-flat major(Dominant), followed by A-flat major(Tonic).

The key specialization of expert 7 is difficult to evaluate quantitatively in Fig. 1. That is because F-sharp major and C-sharp major are the most underrepresented keys in the test set, thus E major occurrences overshadow F-sharp major occurrences. This data-related limitation will be evaluated qualitatively in the next section.

4.4. Qualitative Analysis

To further test these observations, we use piano roll visualizations with a heat map coloring for each note's activation coefficient (Fig. 2). A higher coefficient is highlighted with a lighter color, while a lower coefficient is shown in a darker one. Black notes have a coefficient lower than the threshold (0.4), thus are not highlighted. The green vertical lines are the labels of the ground-truth key changes from the dataset. The scores that we use are Bach's prelude from WTC I No. 23 and chorale No. 343.

We observe that experts activate for specific keys based on dominant-to-tonic relationships, clustering note activations based on the key scales with higher activation on degrees I and V. In Bach's Prelude No. 23, expert 7 shows high coefficients for B major and F-sharp major notes while avoiding G-sharp minor, whereas expert 27 activates exclusively in G-sharp minor, which is enharmonic to A-flat minor. This demonstrates clear key-specific specialization following the circle of fifths.

Experts focus less on the third of the scale, interpreting major and minor modes as parallel keys with higher values on major modes. The activation pattern of expert 7 on B-major to F-sharp

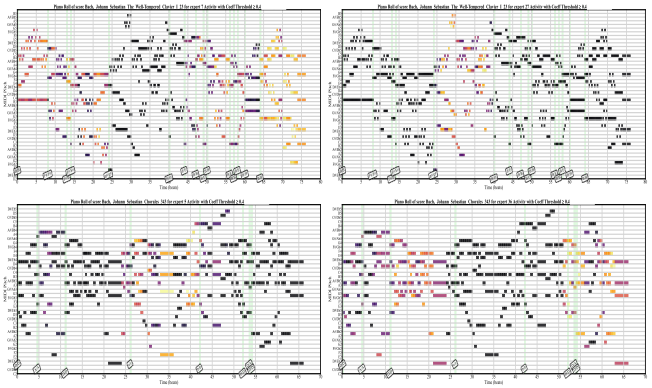


Fig. 2. Piano rolls visualization of expert activations with threshold ≥ 0.4 from layer 12. Notes with a lighter color have a higher coefficient value, while black notes have a coefficient $<$ threshold. Vertical green lines are the ground-truth key changes. **Zoom for clarity.**

major instead of E-major to B-major is highlighted here. In general, every activation of expert 7 in segments of E-major or minor is sparse and does not cluster notes, instead, it only predicts the tonic.

In Chorale 343, experts 5 and 36 both activate on A major, with expert 5 showing fewer activated notes. This supports our finding that experts activate on a key segment depending on whether they work as the tonic or as the dominant in the chord progression. This pattern aligns with Western classical music's fundamental dominant-to-tonic structures.

Analysis of layers 10 and 11 reveals that the experts work as a single unit, specializing in one harmonic function or task. For example, expert 29 predicts only minor 7th notes in dominant major chords, while expert 34, an auxiliary task, identifies harmony onsets. Unfortunately, those specializations are harder to evaluate or visualize.

We chose 48 experts as the optimal count. Fewer experts create polysemantic relations, predicting more than 2 keys, still in a dominant-to-tonic pattern. Experts in the range of 128-512 specialize monosemantically in a dominant-to-tonic pattern but drop the activation for the rest notes of the scale, destroying the clusters and losing the musicological interpretability. Since the model operates on a per-token-to-note paradigm, this behavior is consistent with [13]. However, to our surprise, for more than 1000 experts, one expert specializes in almost every key, losing any interpretative value.

5. CONCLUSION

This paper has introduced μMoE-RNBERT, a novel architecture for roman numeral analysis that combines factorized, multilinear experts into the state-of-the-art RNBERT model. We have shown that this approach is parameter-efficient and, for the first time, interpretable by its nature with minimal losses in accuracy. Our findings show that by replacing MLPs with μMoE layers, the model achieves similar performance in conjunction with predictions that are musicologically interpretable. Finally, for future work, this approach can be applied to different MIR tasks, not only in harmonic but also rhythmical contexts, like melody harmonization, composer classification, and automatic music transcription, to name a few.

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7. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required.

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